# Supplier invoice query handling - Solution Design for Planning

## Executive Summary

Source-grounded planning design using workflow, deterministic rules, the ready read-only invoice-status GET and human judgment. The two-working-day target is acknowledgment. Bank-details POST is prohibited; invoice PATCH is unknown and unavailable. Proposed components and controls, sourced current steps and unanswered technical questions are distinguished. Planning approval does not authorize implementation.

## Summary

Planning design for supplier invoice queries. The source-supported process captures queries, verifies requester authority, identifies invoice references, retrieves status and requires a human to approve and send the reply. Acknowledge within two working days; no resolution SLA is established. The ready read-only GET is supported. Bank-details POST is prohibited; invoice PATCH is unknown and unavailable pending separate evidence and approval. The To-Be workflow and technical controls below are proposals, not verified deployed capabilities. This document is approved only for documentation and planning after independent review.

## Architecture

Proposed planning architecture: workflow orchestration coordinates query capture, acknowledgment within two working days, requester verification, invoice-reference matching, the ready read-only invoice-status GET, human response approval and audit recording. Deterministic rules handle structured references; people resolve ambiguous references, disputes, disclosure authority and unresolved lookup failures. The ready GET is preferred to RPA for this lookup. Document intelligence is an optional alternative for unstructured reference extraction, subject to separate justification and evidence; it is not a required dependency or authority to approve or send replies. Event capture, case management and audit components are proposed design responsibilities, not verified deployed services. The prohibited bank-details POST is excluded. The unknown invoice PATCH remains unavailable pending separate evidence and approval. Deployment topology, endpoint configuration, authentication, capacity and service levels remain open design questions. Planning approval does not authorize implementation.

## Components

Proposed components and responsibilities:
- Query intake and acknowledgment workflow: retain the original message and track the two-working-day acknowledgment target.
- Requester-verification work item: a person checks the supplier contact record before disclosure.

- Reference validation rules: match invoiceId and supplierId exactly; do not infer missing references.
- Read-only API adapter: use the ready invoice-status GET and the documented synthetic error contract.
- Human exception and reply review: investigate ambiguity/disputes, approve and send responses.
- Evidence record: link the original source and approved response; retention duration is unresolved.

These are planning responsibilities, not claims that services or controls are already implemented.

## Interfaces

Supported fictional interface: GET /sandbox/invoices/{invoiceId}/status is ready for read-only lookup. Finance Applications Owner - Accounts Payable is accountable; the application is Stable and status data is Confidential under the separately authored synthetic supplement.

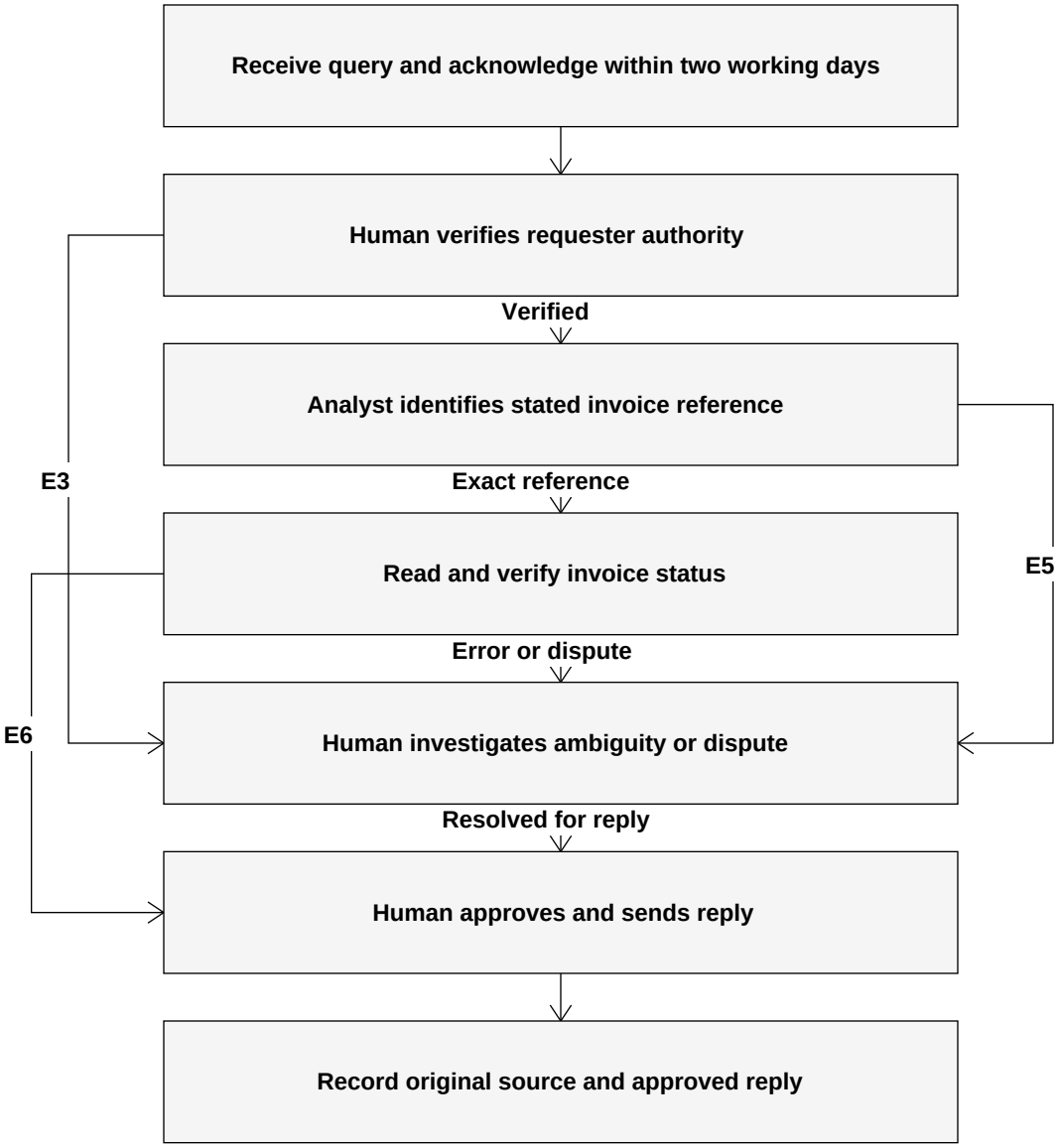
Invalid reference requires correction. Not found routes to human review. Access denied stops lookup and refers the issue to the accountable owner. Temporary failure/timeout uses the existing bounded retry policy where applicable, then human review; no numeric retry limit is supplied. Missing records and technical errors never mean successful status.

POST /sandbox/supplier/bank-details is prohibited and excluded. PATCH /sandbox/invoices/{invoiceId} is unknown and unavailable; neither operation is promoted by the ready GET. These read APIs do not approve, pay, update or cancel anything. Endpoint deployment configuration, authentication, capacity and SLAs remain unknown.

## Data design

Data captured includes supplier query email, received time, requester identity, and invoice references. The original message is linked and acknowledged within two working days. Invoice status data retrieved includes invoiceId, supplierId, and paymentStatus. Data classification is Confidential for invoice status. Human approval is required before sending responses. Retention duration for recorded evidence is an open question.

## As is flow

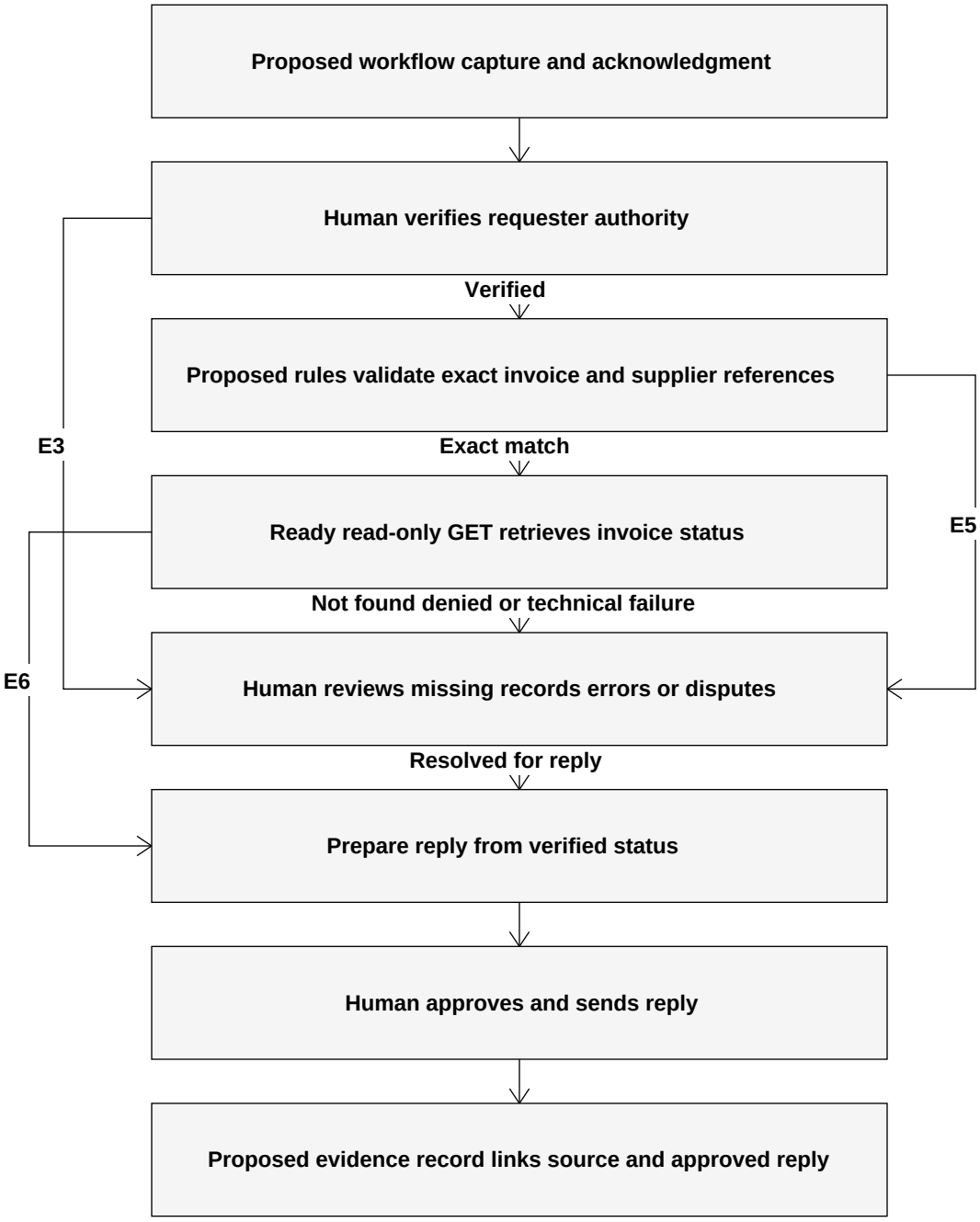



### Transition details

[A] Receive query and acknowledge within two working days - continue -> [B] Human verifies requester authority
[B] Human verifies requester authority - Verified -> [C] Analyst identifies stated invoice reference
[B] Human verifies requester authority - Unresolved -> [E] Human investigates ambiguity or dispute
[C] Analyst identifies stated invoice reference - Exact reference -> [D] Read and verify invoice status
[C] Analyst identifies stated invoice reference - Missing or ambiguous -> [E] Human investigates ambiguity or dispute
[D] Read and verify invoice status - Verified status -> [F] Human approves and sends reply
[D] Read and verify invoice status - Error or dispute -> [E] Human investigates ambiguity or dispute

[E] Human investigates ambiguity or dispute - Resolved for reply -> [F] Human approves and sends reply
[F] Human approves and sends reply - continue -> [G] Record original source and approved reply

## To be flow




### Transition details

[A] Proposed workflow capture and acknowledgment - continue -> [B] Human verifies requester authority
[B] Human verifies requester authority - Verified -> [C] Proposed rules validate exact invoice and supplier references
[B] Human verifies requester authority - Unresolved -> [E] Human reviews missing records errors or disputes
[C] Proposed rules validate exact invoice and supplier references - Exact match -> [D] Ready read-only GET retrieves invoice status
[C] Proposed rules validate exact invoice and supplier references - Invalid or ambiguous -> [E] Human reviews missing records errors or disputes
[D] Ready read-only GET retrieves invoice status - Matching status -> [F] Prepare reply from verified status
[D] Ready read-only GET retrieves invoice status - Not found denied or technical failure -> [E] Human reviews missing records errors or disputes
[E] Human reviews missing records errors or disputes - Resolved for reply -> [F] Prepare reply from verified status
[F] Prepare reply from verified status - continue -> [G] Human approves and sends reply
[G] Human approves and sends reply - continue -> [H] Proposed evidence record links source and approved reply

## Security controls

Source-supported boundaries: verify requester authority before disclosure; require human approval and sending of replies; route disputes and errors to people. Status information is Confidential under the human-authored synthetic supplement.
Proposed controls requiring technical design and verification: least-privilege access, audit correlation, monitoring, segregation of duties and safe recovery for read failures. No control effectiveness or compliance is certified here. No write compensation is assumed because the proposed API path is read-only. Optional AI would require separately justified data-access and safety controls before selection.

## Operational design

Proposed workflow tracks receipt and acknowledgment within two working days, with human verification before status disclosure and human approval before sending. No resolution SLA is established. Validation errors require corrected references; not-found, ambiguous and disputed cases require human review. Access-denied errors stop lookup and go to the accountable owner. Temporary service errors follow the existing bounded retry policy where applicable, then human review. Never convert an error into a successful status. Monitoring, alert ownership, retry configuration, retention and service-level targets remain design questions.

## Design decisions

Planning decisions: prefer the ready read-only GET with workflow and deterministic rules; retain human judgment for requester authority, ambiguity, disputes and outgoing replies. Exclude the prohibited bank-details POST and leave unknown PATCH unavailable. Keep the two-working-day target specifically for acknowledgment. The proposed To-Be flow labels workflow/rules as proposed and preserves human review. Optional AI is not a dependency of this design; it requires separate justification and evidence. Planning approval authorizes document content and discovery planning, not implementation or deployment.

## Open questions and proposals

Selected-design questions: confirm deployment configuration, authentication/least privilege, capacity, service levels, retention, monitoring ownership and the applicable bounded retry settings before dependent implementation. Record answers with source evidence; do not infer them.

Optional-alternative questions: agent controllability, adaptive next steps, incremental value, tool/path selection and ambiguity evidence matter if AI is later proposed; they do not make AI necessary now.

Modernization context: strategic lifespan, technical health and vendor roadmap remain unknown where not established; they inform future investment decisions, not the documented read-only process facts. The synthetic supplement already establishes accountable ownership, Stable operation, Confidential classification and the read error contract; do not relabel those supplied facts as missing.

Discovery tasks may be planned and independently approved. Affected implementation retains its dependencies; planning approval does not certify unresolved facts or implementation readiness.